

Keyang Zheng

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EDUCATION

University of Pittsburgh

School of Computing and Information

Ph.D. in Information Science

Thesis: Improving Human-Agent Collaboration through Effective Gestured Based Communication

Pittsburgh, US

Fall 2018 – May 2026

(Defended on 23 March 2026)

University of Pittsburgh

School of Computing and Information

M.S. in Information Science

Pittsburgh, US

2016 - 2018

Nanjing University

B.E. Computer Software Engineering

Software Institute

Outstanding Student Leadership Award in the Software Institute (2013)

Nanjing, China

2012 - 2016

RESEARCH EXPERIENCE

University of Pittsburgh

Doctoral Researcher, School of Computing and Information

Pittsburgh, US

Sept 2023 – May 2026

- Built instrumentation and data pipelines to capture and investigate team collaboration behaviors in competitive online multiplayer games across 30,000+ game sessions.
- Applied statistical modeling (Clustering Analysis, Structural Equation Models) on log data to model users' communication behaviors.
- Conducted interview sessions to identify user preferences and perceptions of nonverbal communications in virtual environments.
- Designed and implemented a communication framework for AI agents based on identified user models to enhance human-AI teaming experience and effectiveness.
- Designed and conducted user experiments evaluating the communication framework in human-AI teams, measuring impact on user experience and task outcomes.
- Mentored one undergraduate students resulting in one accepted peer-reviewed paper (FDG 2026).
- Published research findings in 4 peer-reviewed papers at top HCI venues (CHI PLAY, ACM Transactions on Social Computing).

Carnegie Mellon University

Research Assistant, Robotic Institute

Pittsburgh, US

Dec 2019 – Aug 2023

- Collaborated with research teams from MIT, ASU, Cornell, USC, etc. in the Artificial Social Intelligence for Successful Teams project, with the aim of developing artificial

agents to assist with human teaming activities using targeted and contextually relevant interventions.

- Modeled users' decision making and collaborative behaviors during a simulated search and rescue mission in Minecraft using analytical methods on game log data.
- Designed user experiments to evaluate users' state of mind and artificial agents' influence on users' performance.
- Designed Theory of Mind based models to identify misbeliefs or information imbalance among users and targeted interventions in team collaboration for the deep reinforcement learning based agents.
- Published research findings in peer-reviewed papers at top venues (IEEE SMC, HFES).

University of Pittsburgh

Pittsburgh, US

Research Assistant, Graduate School of Public Health

Aug 2018 – Apr 2020

- Analyzed and modeled alcohol/substance abuse behaviors in California using survey and interview data.
- Developed a preliminary agent-based simulation model on the community influence of alcohol/substance abuse behaviors.

TEACHING EXPERIENCE

Teaching Fellow

INFSCI 0510 Data Analysis

Spring 2024, Summer 2024, Fall 2025, Spring 2026

University of Pittsburgh

Sole instructor for undergraduate course (40–55 students). Responsible for curriculum design, lecture delivery, assessment setting, grading, and individualized feedback. Course topics include data preprocessing, visualization, clustering analysis, regression, and classification.

INFSCI 0201 Intermediate Programming with Python

Fall 2024, Spring 2025

University of Pittsburgh

Sole instructor for undergraduate course (10–20 students). Responsible for curriculum design, lecture delivery, coding laboratory design, assessment setting, grading, and individualized student feedback, major topics including Object-Oriented Programming and Functional Programming.

TALKS AND PRESENTATIONS

Shared Deliberation in Facebook Support Groups for Sickle Cell Patients and Caregivers

Foundation for Sickle Cell Disease Research, Miami, US

April 2017

Exploration of online health support groups through the lens of sentiment analysis.

iConference 2018, Sheffield, UK

March 2018

Understanding Player's Gesture-Based Communicative Behavior in MOBA Games.

The Annual Symposium on Computer-Human Interaction in Play, Stratford, Canada

Oct 2023

HONORS & AWARDS

Outstanding Student Leadership Award in the Software Institute

2013

SERVICE

Reviewer
CHI 2025
WWW 2018, 2019
CSCW 2021, 2023
CHI Play 2022, 2023, 2025
FDG 2026
IEEE Transaction on Human-Machine Systems

Associate Chair
CHI Play 2025, 2026 (Working in Progress Track)

PUBLICATIONS

Zheng, K., Healy, P., Wang, S., & Farzan, R. (2026). Toxic Pings: An Interview Study on Hostile Nonverbal Communication Among Teammates in DotA 2 and League of Legends. In Proceedings of the Foundations of Digital Games (FDG 2026). (accepted)

Zheng, K. (2026). *Improving Human-Agent Collaboration Through Effective Gestured Based Communication* (Doctoral dissertation, University of Pittsburgh).

Zheng, K., Li, A., & Farzan, R. (2018). Exploration of online health support groups through the lens of sentiment analysis. In Transforming Digital Worlds: 13th International Conference, iConference 2018, Sheffield, UK, March 25-28, 2018, Proceedings 13 (pp. 145-151). Springer International Publishing.

Li, H., Zheng, K., Lewis, M., Hughes, D., & Sycara, K. (2021, September). Human theory of mind inference in search and rescue tasks. In Proceedings of the Human Factors and Ergonomics Society Annual Meeting (Vol. 65, No. 1, pp. 648-652). Sage CA: Los Angeles, CA: SAGE Publications.

Li, H., Le, L., Chis, M., Zheng, K., Hughes, D., Lewis, M., & Sycara, K. (2023, January). Sequential theory of mind modeling in team search and rescue tasks. In Computational Theory of Mind for Human-Machine Teams: First International Symposium, ToM for Teams 2021, Virtual Event, November 4–6, 2021, Revised Selected Papers (pp. 158-172). Cham: Springer Nature Switzerland.

Zheng, K., Stein, B., & Farzan, R. (2023). Use Ping Wisely: A Study of Team Communication and Performance under Lean Affordance. *ACM Transactions on Social Computing*, 5(1-4), 1-26.

Zheng, K., & Farzan, R. (2023). Understanding Player's Gesture-Based Communicative Behavior in MOBA Games. *Proceedings of the ACM on Human-Computer Interaction*, 7(CHI PLAY), 1068-1090.

Chis, M., Li, H., Zheng, K., Lewis, M., Hughes, D., & Sycara, K. (2023, October). The Cognitive Load–Productivity Tradeoff in Task Switching. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (p. 21695067231193677). Sage CA: Los Angeles, CA: SAGE Publications.

Li, H., Chis, M., Zheng, K., Lewis, M., Hughes, D., & Sycara, K. (2023, September). Sentiment analysis of Artificial Advisors in Search and Rescue Tasks. In *Proceedings of the Human Factors and Ergonomics Society Annual Meeting* (Vol. 67, No. 1, pp. 2564-2570). Sage CA: Los Angeles, CA: SAGE Publications.

Li, H., Fan, Y., Zheng, K., Lewis, M., & Sycara, K. (2023, October). Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 4865-4870). IEEE.

Hughes, D., Li, H., Chis, M., Oguntola, I., Stepputtis, S., Zheng, K., Campbell, J., Sycara, K. and Lewis, M. (2023, October). A Framework for Intervention Based Team Support in Time Critical Tasks. In *2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC)* (pp. 4805-4812). IEEE.

Zheng, K., Li, H., Sycara, K., & Lewis, M. (2021). Human Theory of Mind Inference in Search and Rescue Tasks.